

Specimen & Microscope

Thin blood smear
100x oil immersion
Compound optics



Camera & Interface

Raspberry Pi HQ
12.3 MP IMX477
CSI-2 / C-mount



Edge Compute Platform

Raspberry Pi 4B
Quad Cortex-A72
Linux 64-bit / 4GB



Embedded Inference

YOLOv8n (INT8)
ONNX Runtime
~245 ms / frame



Clinical Triage

RBC / WBC / PLT
CLSI EP09c metrics
PDF triage report